

A Dirichlet Pólya Inequality for Three-Dimensional Spherical Shells

Dependency-compressed review draft

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Scope and status. This document reorganizes an internally validated proof into a conventional review manuscript. It proves a theorem for the class of genuine three-dimensional spherical shells. It does not claim the general Pólya conjecture, literature novelty, priority, or publication readiness. The final theorem assembly and the non-tiling argument are written out. The long compact-middle argument is presented as a sequence of precise lemmas with exact domains, seams, constants, and pointers to the audited detailed proofs. A journal submission should move those detailed arguments and the computer-assisted contracts into conventional appendices rather than cite repository workflow records.

Abstract

For $0 < r < R$, let

$$A_{r,R} = \{x \in \mathbb{R}^3 : r < |x| < R\}.$$

We prove, with the strict Dirichlet counting convention, that

$$N_D(A_{r,R}, \Lambda) \leq L_3 |A_{r,R}| \Lambda^{3/2}, \quad L_3 = \frac{1}{6\pi^2}, \quad \Lambda \geq 0.$$

The proof separates variables, converts the radial spectrum into a strict multiplicity-weighted phase count, and combines analytic endpoint estimates, finite angular and radial mode inventories, and a rigorously certified final compact window. We also give an elementary proof that no genuine spherical shell tiles $\mathbb{R}^3$ by congruent rigid-motion copies with disjoint interiors, under either exact or almost-everywhere coverage.

1 Main results and conventions

The eigenvalues of the Dirichlet Laplacian on a bounded domain Ω are listed with multiplicity as

$$0 < \lambda_1^D(\Omega) \leq \lambda_2^D(\Omega) \leq \dots$$

Throughout we use the strict counting function

$$N_D(\Omega, \Lambda) := \#\{j : \lambda_j^D(\Omega) < \Lambda\}. \tag{1.1}$$

This convention matters at every spectral wall. We write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

Theorem 1.1 (Spectral inequality). *For every $0 < r < R$ and every $\Lambda \geq 0$,*

$$\boxed{N_D(A_{r,R}, \Lambda) \leq \frac{2}{9\pi} (R^3 - r^3) \Lambda^{3/2} = L_3 |A_{r,R}| \Lambda^{3/2}, \quad L_3 = \frac{1}{6\pi^2}.} \tag{1.2}$$

Theorem 1.2 (Non-tiling). *For every $0 < r < R$, no family of congruent rigid-motion copies of $A_{r,R}$ has pairwise-disjoint interiors and covers $\mathbb{R}^3$ exactly or up to a three-dimensional Lebesgue-null set. The same conclusion holds when coverage is formulated using closed shells.*

The two theorems concern literally the same complete shell family. Theorem 1.2 is logically independent of the spectral proof.

2 Separation of variables and strict phase counting

We first normalize the outer radius to one and write

$$A_{\rho,1} = \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \quad 0 < \rho < 1.$$

Proposition 2.1 (Exact separated spectrum). *Under spherical-harmonic decomposition and the unitary radial substitution $u(r) = rR(r)$, the Dirichlet Laplacian is the orthogonal sum*

$$-\Delta_{A_{\rho,1}}^D \simeq \bigoplus_{\ell=0}^{\infty} \bigoplus_{m=-\ell}^{\ell} L_{\ell}, \quad L_{\ell} = -\frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{r^2}, \quad (2.1)$$

where L_{ℓ} acts on $H^2(\rho, 1) \cap H_0^1(\rho, 1)$. Each radial block has a positive, simple, complete spectrum. The full resolvent is compact because

$$\|(L_{\ell} + 1)^{-1}\| \leq \frac{1}{1 + \ell(\ell+1)} \rightarrow 0. \quad (2.2)$$

Put $\nu = \ell + \frac{1}{2}$. The positive eigenfrequencies of the ℓ -th block are exactly the positive zeros of

$$F_{\nu,\rho}(k) = J_{\nu}(\rho k)Y_{\nu}(k) - Y_{\nu}(\rho k)J_{\nu}(k). \quad (2.3)$$

For $K > 0$, choose the continuous Bessel phase by

$$J_{\nu}(x) + iY_{\nu}(x) = M_{\nu}(x)e^{i\theta_{\nu}(x)}, \quad \theta_{\nu}(0+) = -\frac{\pi}{2},$$

and define

$$\Psi_{\nu,\rho}(K) = \theta_{\nu}(K) - \theta_{\nu}(\rho K), \quad \gamma_{\rho,K}(\nu) = \frac{\Psi_{\nu,\rho}(K)}{\pi}. \quad (2.4)$$

The function $K \mapsto \Psi_{\nu,\rho}(K)$ is strictly increasing from zero to infinity, and

$$F_{\nu,\rho}(K) = 0 \iff \Psi_{\nu,\rho}(K) \in \pi\mathbb{N}. \quad (2.5)$$

Consequently,

$$N_D(A_{\rho,1}, K^2) = \sum_{\nu_{\ell} < K} 2\nu_{\ell}[\gamma_{\rho,K}(\nu_{\ell})]_{<}, \quad \nu_{\ell} = \ell + \frac{1}{2}, \quad (2.6)$$

where

$$[x]_{<} := \#\{n \in \mathbb{N} : n < x\} = \max\{0, [x] - 1\}. \quad (2.7)$$

In particular, $[m]_{<} = m - 1$ when $m \in \mathbb{N}$. If several angular channels share an eigenfrequency, all equal-threshold modes are excluded at that wall and enter above it with the sum of their multiplicities.

The phase convention and special-function identities used here are standard; see [5, 2]. The exact reconstruction statement is `state/lemma_packets/SHELL-sturm-liouville-completeness.md`. The accepted proof evidence consists of `rounds/polya-main/round_004/responses/sturm-liouville-incumbent.md`, `rounds/polya-main/round_004/reviews/sturm-liouville-clean-room.md`, and `rounds/polya-main/round_004/reviews/sturm-liouville-adversarial.md`.

Proposition 2.2 (Phase-to-lattice reduction). *Let $K > 0$. For $a > 0$, set*

$$G_a(z) = \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), \quad 0 \leq z \leq a, \quad (2.8)$$

and $G_a(z) = 0$ for $z > a$. Put $\mu = \rho K$ and

$$A_{\rho,K}(z) = G_K(z) - G_\mu(z). \quad (2.9)$$

For $0 \leq z < \mu$, define

$$H_\mu(z) = \frac{3\mu^2 + 2z^2}{24\pi(\mu^2 - z^2)^{3/2}},$$

and define the auxiliary function—denoted F_μ in the annulus source, but written $\mathcal{F}_\mu$ here to distinguish it from the Bessel determinant—by

$$\mathcal{F}_\mu(z) = \begin{cases} \max\{G_\mu(z) - H_\mu(z), -1/4\}, & 0 \leq z < \mu, \\ -1/4, & z \geq \mu. \end{cases}$$

Finally, $\Phi_{K,\mu}(z) = G_K(z) - G_\mu(z) = A_{\rho,K}(z)$ on $0 \leq z \leq \mu$.

The global phase enclosure [3, Theorem 5.1 and Lemma 5.2] gives, for $0 \leq \nu \leq K$,

$$0 < \gamma_{\rho,K}(\nu) < G_K(\nu) - \mathcal{F}_\mu(\nu) \leq G_K(\nu) + \frac{1}{4}, \quad (2.10)$$

and, when $\nu \leq \mu$, the sharper two-sided action enclosure

$$\Phi_{K,\mu}(\nu) < \gamma_{\rho,K}(\nu) < \Phi_{K,\mu}(\nu) + \frac{1}{4}. \quad (2.11)$$

Set

$$s_\mu(\nu) = \begin{cases} \min\{H_\mu(\nu), 1/4\}, & 0 \leq \nu < \mu, \\ 1/4, & \mu \leq \nu \leq K, \end{cases} \quad P_{\rho,K}(\nu) = A_{\rho,K}(\nu) + s_\mu(\nu).$$

Combining the two phase bounds gives

$$\gamma_{\rho,K}(\nu) < P_{\rho,K}(\nu) \leq A_{\rho,K}(\nu) + \frac{1}{4}.$$

Thus the strict spectral count is bounded by the optimized proxy, and in particular by the coarse proxy through the explicit chain

$$N_D(A_{\rho,1}, K^2) \leq \sum_{\substack{\ell \geq 0 \\ \nu_\ell < K}} 2\nu_\ell [P_{\rho,K}(\nu_\ell)]_< \leq S_{\rho,K}.$$

Here

$$S_{\rho,K} = \sum_{\substack{\ell \geq 0 \\ \nu_\ell < K}} 2\nu_\ell \left[A_{\rho,K}(\nu_\ell) + \frac{1}{4} \right]_<. \quad (2.12)$$

The continuous target is exact [1]:

$$\int_0^\infty 2z A_{\rho,K}(z) dz = \frac{2}{9\pi} (1 - \rho^3) K^3. \quad (2.13)$$

The remaining task is to show that strict floor losses, phase slack, and the half-integer multiplicity mesh do not exceed the target in (2.13). The exact reconstruction contracts are `state/lemma_packets/SHELL-phase-enclosures.md` and `state/lemma_packets/SHELL-weighted-lattice-fractional.md`. Accepted proof evidence is supplied by the Round 2 phase incumbent, clean-room proof, and adversarial review, and by the corresponding Round 3 low-interface records:

Internal proof record. rounds/polya-main/round_002/responses/phase-enclosure-incumbent.md; rounds/polya-main/round_002/reviews/phase-enclosure-clean-room.md; rounds/polya-main/round_002/reviews/phase-enclosure-adversarial.md; rounds/polya-main/round_003/responses/low-interface-fixed-rho-incumbent.md; rounds/polya-main/round_003/reviews/low-interface-fixed-rho-clean-room.md; rounds/polya-main/round_003/reviews/low-interface-fixed-rho-adversarial.md.

For later use, write

$$W(\rho, K) = \frac{2}{9\pi}(1 - \rho^3)K^3. \quad (2.14)$$

3 The two ratio endpoints

Define

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \quad C_* = \frac{1}{2} + \frac{8\sqrt{2}}{15}, \quad \rho_* = \frac{\omega_0}{1 + 2C_*}. \quad (3.1)$$

Lemma 3.1 (Small-hole endpoint). *For $0 < \rho \leq \rho_*$ and $K \geq 0$,*

$$N_D(A_{\rho,1}, K^2) \leq W(\rho, K). \quad (3.2)$$

Proof. The multiplicity identity rewrites (2.12) as a sum of shifted tails, one for each $r \in \mathbb{N}_0$ and beginning at $x_r = r + \frac{1}{2}$. The ordinary-floor tail majorant is legitimate because $[x]_< \leq [x]$ for every $x \geq 0$. Every tail beginning at or above the inner interface $\mu = \rho K$ is controlled by the convex high-angular theorem of [1].

The separately proved low-interface estimate, using the concave and convex floor-sum estimates from [3], controls a tail beginning below μ whenever

$$0 < \rho < \omega_0, \quad K(\omega_0 - \rho) \geq C_*. \quad (3.3)$$

If $\mu \leq \frac{1}{2}$, then every $x_r \geq \frac{1}{2} \geq \mu$, so only the high-angular case occurs. If $\mu > \frac{1}{2}$ and $0 < \rho \leq \rho_*$, then

$$K(\omega_0 - \rho) > \frac{\omega_0 - \rho}{2\rho} \geq \frac{\omega_0 - \rho_*}{2\rho_*} = C_*. \quad (3.4)$$

Hence every tail is controlled. Summing the tail integrals gives the target bound corresponding to (2.13), which proves (3.2). The face $\rho = \rho_*$ is included. At $K = 0$, both sides vanish. $\square$

The exact statement and shifted-tail inequalities are in state/lemma_packets/SHELL-rho-zero-endpoint.md. The proof and its independent checks are rounds/polya-main/round_007/responses/small-hole-incumbent.md, rounds/polya-main/round_007/reviews/small-hole-clean-room.md, and rounds/polya-main/round_007/reviews/small-hole-adversarial.md.

Lemma 3.2 (Thin-shell endpoint). *For $7/8 \leq \rho < 1$ and $K \geq 0$,*

$$N_D(A_{\rho,1}, K^2) \leq W(\rho, K). \quad (3.5)$$

Proof outline with exact closing margins. Put $\varepsilon = 1 - \rho \in (0, 1/8]$ and $a = \varepsilon K$. The proof has two inclusive pieces.

On $0 \leq a \leq \pi/(4\varepsilon)$, the radial form comparison has the correct upper-count direction. Strict radial counting gives the number of active one-dimensional radial levels as

$$M = \max \left\{ 0, \left\lceil \frac{a}{\pi} \right\rceil - 1 \right\}.$$

Define

$$I(a) = \frac{2a^3}{3\pi}, \quad S_0(a) = \sum_{n=1}^M (a^2 - n^2\pi^2), \quad D(a) = I(a) - S_0(a).$$

For $a > \pi$, write $a/\pi = M + t$, where $M \geq 1$ and $0 < t \leq 1$, using $(M, t) = (m - 1, 1)$ at $a = m\pi$. The exact product deficit satisfies

$$\frac{D(a)}{\pi^2} = \frac{M^2}{2} + M \left(t^2 + \frac{1}{6} \right) + \frac{2t^3}{3}, \quad D(a) > \frac{2}{5}a^2. \quad (3.6)$$

The angular-ceiling and quarter-disk cost is at most

$$\left(\frac{1}{6} + \frac{\varepsilon}{4} + \frac{\varepsilon^2}{9} \right) a^2.$$

At $\varepsilon = 1/8$, the remaining reserve is

$$\frac{2}{5} - \left(\frac{1}{6} + \frac{1}{32} + \frac{1}{576} \right) = \frac{577}{2880} > 0. \quad (3.7)$$

On $a \geq \pi/(4\varepsilon)$, let $\mathcal{A} : [0, a] \rightarrow [0, T]$, $T = a/\pi$, be the exact decreasing shell action from the phase reduction, let $R_{\mathcal{A}} : [0, T] \rightarrow [0, a]$ be its decreasing inverse, and put $F_{\mathcal{A}} = R_{\mathcal{A}}^2$ and $g = -F'_{\mathcal{A}}$. Splitting at the actual ungridded curvature interface, shifted-sawtooth Stieltjes integration gives the radial deficit

$$D_{\text{rad}} := \int_0^T F_{\mathcal{A}}(t) dt - \sum_{\substack{n \geq 1 \\ n-1/4 < T}} F_{\mathcal{A}}(n - 1/4)$$

and the bound

$$D_{\text{rad}} \geq \frac{\rho^2 a^2}{4} - \frac{\pi \rho a}{4}, \quad (3.8)$$

whereas strict half-integer angular counting costs at most

$$E = \left(\frac{a}{\pi} + \frac{1}{4} \right) \left(\varepsilon a + \frac{\varepsilon^2}{4} \right). \quad (3.9)$$

The relevant lower screen decreases and the upper screen increases on $0 < \varepsilon \leq 1/8$. At the closed endpoint,

$$\frac{D_{\text{rad}}}{a^2} \geq \frac{21}{256}, \quad \frac{E}{a^2} < \frac{193}{4096},$$

so

$$D_{\text{rad}} - E > \frac{143}{4096} a^2 > 0. \quad (3.10)$$

Both arguments include the common face $a = \pi/(4\varepsilon)$, all radial and angular integer walls, and the face $\rho = 7/8$. Together they cover every $a \geq 0$ and prove (3.5). $\square$

The frozen exact statement is `state/lemma_packets/SHELL-rho-one-endpoint-round16.md`; that packet is a contract, not the proof. The complete argument is in `rounds/polya-main/round_016/responses/thin-endpoint-incumbent.md`, with the independent reconstruction `rounds/polya-main/round_016/reviews/thin-endpoint-clean-room.md` and adversarial audit `rounds/polya-main/round_016/reviews/thin-endpoint-adversarial-referee.md`.

4 The compact-middle theorem

This section compresses the longest part of the proof into exact regional lemmas. Each lemma has an analytic proof or a rigorously certified finite part in the listed repository record.

Set

$$\begin{aligned} \rho_0 &= \frac{1}{\sqrt{337}}, & \rho_c &= \frac{1}{1+2\pi}, & z_\rho &= \frac{\pi}{1-\rho}, \\ k_m(\rho) &= \sqrt{z_\rho^2 + m(m+1)} \quad (m \in \mathbb{N}_0), & d &= \frac{\sqrt{337}}{2}. \end{aligned} \quad (4.1)$$

To make the upper-frequency wall explicit, set

$$\begin{aligned} C_0 &= 1 + \frac{8\sqrt{2}}{15}, & a_\rho &= \frac{2\pi\rho}{1-\rho}, & \eta_\rho &= G_1\left(\max\left\{\rho, \frac{1}{2}\right\}\right), \\ K_0(\rho) &= \left(\frac{\sqrt{a_\rho} + \sqrt{a_\rho + 4\eta_\rho C_0}}{2\eta_\rho}\right)^2, & H_0(\rho) &= \frac{C_*}{\omega_0 - \rho} \quad (\rho < \omega_0), \end{aligned}$$

and, equivalently to taking the minimum of all active upper owners,

$$U(\rho) = \begin{cases} \min\{K_0(\rho), H_0(\rho)\}, & 0 < \rho < \omega_0, \\ K_0(\rho), & \omega_0 \leq \rho < 5/6, \\ \min\{K_0(\rho), 54/(1-\rho)^2\}, & 5/6 \leq \rho < 1. \end{cases}$$

Thus U is precisely the upper-frequency owner obtained from the fixed-ratio, small-hole, and seam high-energy thresholds. On the final live interval $\rho_c \leq \rho < 39/50$, it is exactly $K_0(\rho)$.

Lemma 4.1 (Primitive compact envelope). *The zero-count, small-hole, fixed-ratio high-energy, and thin-shell estimates cover every compact-middle point except the exact nonrectangular set*

$$\mathcal{D}_{16} = \left\{ \rho_* < \rho < \frac{7}{8} : L(\rho) < K < U(\rho) \right\}, \quad (4.2)$$

where

$$L(\rho) = \max\left\{\frac{1}{2\rho}, \frac{\pi}{1-\rho}\right\}.$$

In particular, the accepted analytic envelope already gives the global high-frequency theorem

$$0 < \rho < 1, \quad K \geq 295^2 \implies N_D(A_{\rho,1}, K^2) \leq W(\rho, K). \quad (4.3)$$

The precursor estimates are collected in `state/lemma_packets/SHELL-rho-compact.md`. The exact formula (4.2), including its open faces and the definition of U , is proved in `state/lemma_packets/SHELL-rho-compact-round17.md`.

Lemma 4.2 (Finite angular and radial staircase). *The strict inequality $N_D(A_{\rho,1}, K^2) < W(\rho, K)$ holds on each of the following regions:*

1. $\rho_c \leq \rho \leq 7/8$ and $z_\rho \leq K \leq k_2(\rho)$;
2. $\rho_c \leq \rho \leq 7/8$ and $k_2(\rho) < K \leq k_5(\rho)$;
3. $\rho_0 < \rho < \rho_c$ and $1/(2\rho) < K \leq d$;
4. $\rho_c \leq \rho \leq 7/8$ and $k_5(\rho) < K \leq k_6(\rho)$.

They follow from the one-dimensional lower bound

$$q_{\ell,n}^2 \geq n^2 z_\rho^2 + \ell(\ell+1), \quad (4.4)$$

where $q_{\ell,n}$ is the n -th positive radial eigenfrequency in angular channel ℓ , together with fixed-channel shell-to-ball min-max comparison, exact Bessel-zero inequalities, exhaustive multiplicity inventories, and strict Weyl payments at every fixed or moving entry wall. The external first-positive-zero inputs are from [4]; their conversion to shell exclusions, all higher-zero bounds, and all mode inventories are internal. The corresponding residual is

$$\begin{aligned} \mathcal{D}_{19} = & \left\{ \rho_* < \rho \leq \rho_0, \frac{1}{2\rho} < K < U(\rho) \right\} \\ & \cup \{ \rho_0 < \rho < \rho_c, d < K < U(\rho) \} \\ & \cup \left\{ \rho_c \leq \rho < \frac{7}{8}, k_6(\rho) < K < U(\rho) \right\}. \end{aligned} \quad (4.5)$$

At $\rho = \rho_0$, the candidate lower fiber is empty because $1/(2\rho_0) = d$. All frequency inequalities in (4.5) are strict.

The proof-free packets `state/lemma_packets/SHELL-first-angular-bands-round17-claim.md`, `state/lemma_packets/SHELL-next-angular-staircase-round18-claim.md`, and `state/lemma_packets/SHELL-two-sided-staircase-round19-claim.md` fix the exact statements; they are not proofs. The principal accepted proof records are:

Internal proof record. Round 17: `rounds/polya-main/round_017/responses/analytic-compression-incumbent.md`; `rounds/polya-main/round_017/reviews/analytic-compression-clean-room.md`; `rounds/polya-main/round_017/reviews/analytic-compression-adversarial-referee.md`.

Round 18: `rounds/polya-main/round_018/responses/angular-staircase-incumbent.md`; `rounds/polya-main/round_018/reviews/angular-staircase-clean-room.md`; `rounds/polya-main/round_018/reviews/angular-staircase-constants.md`; `rounds/polya-main/round_018/reviews/angular-staircase-cross-comparison.md`; `rounds/polya-main/round_018/reviews/angular-staircase-adversarial-referee.md`.

Round 19: `rounds/polya-main/round_019/responses/high-ratio-k6-incumbent.md`; `rounds/polya-main/round_019/exploration/lower-ratio-route.md`; `rounds/polya-main/round_019/reviews/two-sided-staircase-clean-room.md`; `rounds/polya-main/round_019/reviews/two-sided-staircase-constants.md`; `rounds/polya-main/round_019/reviews/two-sided-staircase-adversarial-referee.md`.

Lemma 4.3 (Combined lower, high-staircase, and optical closure). *The inequality is strict on both lower components of (4.5) and on*

$$\rho_c \leq \rho \leq \frac{7}{8}, \quad z_\rho \leq K \leq k_{11}(\rho). \quad (4.6)$$

It also holds for every frequency on

$$\frac{39}{50} \leq \rho < 1, \quad K \geq 0, \quad (4.7)$$

and is strict there when $K > 0$. Exact subtraction leaves

$$\mathcal{D}_{20} = \left\{ \rho_c \leq \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho) = K_0(\rho) \right\}. \quad (4.8)$$

The face $\rho = 39/50$ belongs to (4.7), $K = k_{11}(\rho)$ belongs to (4.6), and $K = U(\rho)$ belongs to the inherited high-frequency theorem.

The exact statement is fixed by the proof-free `state/lemma_packets/SHELL-combined-closure-round20-claim.md`. The Round 20 high- and lower-route notes are exploratory precursors, not promoted evidence. The complete accepted evidence, including the conservative cap-74 cell, is

Internal proof record. `rounds/polya-main/round_020/reviews/combined-closure-clean-room.md`; `rounds/polya-main/round_020/reviews/combined-closure-clean-room-addendum.md`; `rounds/polya-main/round_020/reviews/combined-closure-cross-comparison.md`; `rounds/polya-main/round_020/reviews/combined-closure-adversarial-referee.md`.

Lemma 4.4 (Exact closure of the final residual). *Two final estimates hold:*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50}, \quad 12 \leq K \leq 200 \implies N_D(A_{\rho,1}, K^2) < W(\rho, K), \quad (4.9)$$

and

$$\rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq 200 \implies N_D(A_{\rho,1}, K^2) < W(\rho, K). \quad (4.10)$$

The exact guards are

$$\frac{7}{51} < \rho_c, \quad k_{11}(\rho) > 12 \quad (\rho_c \leq \rho < 1). \quad (4.11)$$

The restriction $\rho < 1$ in the second guard is essential. Define

$$\mathcal{C}_{21} = \mathcal{D}_{20} \cap \{K \leq 200\}, \quad \mathcal{T}_{21} = \mathcal{D}_{20} \cap \{K > 200\}. \quad (4.12)$$

Every point of $\mathcal{C}_{21}$ lies in the rectangle (4.9), and every point of $\mathcal{T}_{21}$ lies in (4.10). The sets are disjoint and exhaust $\mathcal{D}_{20}$, with $K = 200$ assigned only to the compact subtraction owner. Defining

$$\mathcal{D}_{21} := \mathcal{D}_{20} \setminus (\mathcal{C}_{21} \cup \mathcal{T}_{21}),$$

we therefore have

$$\boxed{\mathcal{D}_{21} = \emptyset.} \quad (4.13)$$

The exact scoped statement is `state/lemma_packets/SHELL-exact-d20-closure-round21-scoped.md`, which is not by itself a proof. Accepted evidence consists of

Internal proof record. `rounds/polya-main/round_021/exploration/exact-d20-closure.md`; `rounds/polya-main/round_021/reviews/exact-d20-closure-clean-room.md`; `rounds/polya-main/round_021/reviews/exact-d20-closure-clean-room-domain-addendum.md`; `rounds/polya-main/round_021/reviews/exact-d20-closure-certificates-source-exec-replacement.md`; `rounds/polya-main/round_021/reviews/exact-d20-closure-adversarial-referee-source-exec.md`.

Certificate boundary for Lemma 4.4

The finite part of (4.9) is an outward-rounded Arb ball-arithmetic partition into 10,580 exact rational leaves. It certifies strict proxy inequalities, interfaces, integer walls, corners, and exact coverage only on

$$[7/51, 39/50] \times [12, 200].$$

It does not certify Bessel roots; the spectral transfer is Proposition 2.2.

For (4.10), 1,286 outward-rounded Arb boxes certify only the base predicates at $K = 200$. Let $\mathcal{B}$ be the aggregate reserve, write $\mathcal{B}_K = \partial_K \mathcal{B}$ and $\mathcal{B}_{KK} = \partial_K^2 \mathcal{B}$, and let F be its one-variable curvature lower screen, with the exact definitions (A13)–(A22) in the listed aggregate-tail contract. On $7/51 \leq \rho \leq 39/50$, the boxes prove precisely

$$\mathcal{B}(\rho, 200) > 0, \quad \mathcal{B}_K(\rho, 200) > 0, \quad F(\rho) > 0,$$

together with the stated domain guards. The universal conclusion for $K > 200$ is analytic: on the same ratio interval and for every $K \geq 200$,

$$\mathcal{B}_{KK}(\rho, K) > F(\rho) > 0,$$

and hence

$$\begin{aligned}\mathcal{B}_K(\rho, K) &= \mathcal{B}_K(\rho, 200) + \int_{200}^K \mathcal{B}_{KK}(\rho, s) ds > 0, \\ \mathcal{B}(\rho, K) &= \mathcal{B}(\rho, 200) + \int_{200}^K \mathcal{B}_K(\rho, s) ds > 0.\end{aligned}\tag{4.14}$$

The two integration premises $\mathcal{B}(\rho, 200) > 0$ and $\mathcal{B}_K(\rho, 200) > 0$ make these integrations strict, while $F(\rho) > 0$ supplies the curvature lower bound; positivity of $\mathcal{B}_{KK}$ alone would not suffice. Thus no finite-box decision is extrapolated beyond its certified domain. The exact contracts are `state/certificate_contracts/ROUND21-compact-proxy-contract.md` and `state/certificate_contracts/ROUND21-aggregate-tail-contract.md`.

Proposition 4.5 (Compact-middle theorem). *For every $\rho_* < \rho < 7/8$ and $K \geq 0$,*

$$N_D(A_{\rho,1}, K^2) \leq W(\rho, K).\tag{4.15}$$

Proof. The primitive cover of Lemma 4.1, the successive closed staircase regions of Lemma 4.2, and the three regions of Lemma 4.3 leave exactly $\mathcal{D}_{20}$. Lemma 4.4 partitions and closes that entire final residual. Equation (4.13) says that no uncovered point remains. All shared faces have a unique declared subtraction owner, while overlapping theorem domains merely give redundant proofs. This proves (4.15). $\square$

5 Proof of the spectral theorem

We now assemble the regional results without importing any new estimate. First, elementary bounds give

$$3\sqrt{3} > 5 > \pi,$$

and hence $\omega_0 > 0$. Also $\sqrt{3} < 2$ and $\pi > 3$ give $\omega_0 < 1/6$. Since $C_* > 0$,

$$0 < \rho_* < \omega_0 < \frac{1}{6} < \frac{7}{8}.\tag{5.1}$$

Therefore

$$(0, 1) = (0, \rho_*] \dot{\cup} (\rho_*, 7/8) \dot{\cup} [7/8, 1).\tag{5.2}$$

Fix $0 < \rho < 1$. At $K = 0$, positivity in Proposition 2.1 gives

$$N_D(A_{\rho,1}, 0) = 0 = W(\rho, 0).\tag{5.3}$$

Let $K > 0$. If $0 < \rho \leq \rho_*$, apply Lemma 3.1. If $\rho_* < \rho < 7/8$, apply Proposition 4.5. If $7/8 \leq \rho < 1$, apply Lemma 3.2. The three cases in (5.2) are disjoint and exhaustive, so

$$N_D(A_{\rho,1}, K^2) \leq \frac{2}{9\pi}(1 - \rho^3)K^3 \quad (K \geq 0).\tag{5.4}$$

Now

$$\omega_3 = \frac{4\pi}{3}, \quad L_3 = \frac{\omega_3}{(2\pi)^3} = \frac{1}{6\pi^2}, \quad |A_{\rho,1}| = \frac{4\pi}{3}(1 - \rho^3),$$

so

$$L_3|A_{\rho,1}| = \frac{2}{9\pi}(1 - \rho^3).\tag{5.5}$$

Taking $K = \sqrt{\Lambda}$ in (5.4) proves the unit-shell form of Theorem 1.1 for every $\Lambda \geq 0$.

For arbitrary $0 < r < R$, put $\rho = r/R$. The unitary dilation

$$(D_R u)(x) = R^{-3/2} u(x/R)$$

maps $L^2(A_{\rho,1})$ onto $L^2(A_{r,R})$, and its Dirichlet form scales by R^{-2} . Hence

$$\lambda_j^D(A_{r,R}) = R^{-2} \lambda_j^D(A_{\rho,1}),$$

and the strict threshold transforms as

$$N_D(A_{r,R}, \Lambda) = N_D(A_{\rho,1}, R^2 \Lambda). \quad (5.6)$$

Applying the unit-shell result at $R^2 \Lambda$,

$$\begin{aligned} N_D(A_{r,R}, \Lambda) &\leq \frac{2}{9\pi} (1 - \rho^3) (R^2 \Lambda)^{3/2} \\ &= \frac{2}{9\pi} (R^3 - r^3) \Lambda^{3/2} \\ &= L_3 |A_{r,R}| \Lambda^{3/2}. \end{aligned}$$

This proves Theorem 1.1 for every $R > 0$, including $R < 1$.

6 Proof that spherical shells do not tile

We prove Theorem 1.2 independently of the spectral argument. Set $\alpha = r/R \in (0, 1)$ and scale by $1/R$. It suffices to consider

$$T = A_{\alpha,1} = \{x : \alpha < |x| < 1\}.$$

Every rigid motion has the form $x \mapsto Qx + a$, with Q orthogonal. Because T is radial, $Q(T) = T$, so every congruent copy is a translate

$$T_a = a + T = \{x : \alpha < |x - a| < 1\}.$$

Assume for contradiction that $\{T_a : a \in \mathfrak{C}\}$ has pairwise-disjoint interiors and covers almost everywhere. Duplicate centers are impossible because they give the same nonempty open tile.

Fix a unit vector e and put

$$c = \frac{1 + \alpha}{2} e, \quad \delta = \frac{1 - \alpha}{2}.$$

Write $B(x, s) = \{y \in \mathbb{R}^3 : |y - x| < s\}$ for the open Euclidean ball. Then $B(c, \delta) \subset T$. Therefore T_a contains $B(a + c, \delta)$, and disjointness implies

$$|a - b| \geq 1 - \alpha \quad (a \neq b). \quad (6.1)$$

Thus the centers are locally finite: a bounded set cannot contain infinitely many points with fixed positive separation. They are also countable, since

$$\mathfrak{C} = \bigcup_{m \geq 1} (\mathfrak{C} \cap \overline{B}(0, m))$$

is a countable union of finite sets.

Each tile boundary is a union of two spheres and is three-dimensional null. The countable union

$$Z = \bigcup_{a \in \mathfrak{C}} \partial T_a$$

is therefore null. Exact open-shell coverage is already a special case of almost-everywhere coverage. If closed shells cover exactly or almost everywhere, removing Z shows that their open interiors still cover almost everywhere. Hence it is enough to assume

$$\left| \mathbb{R}^3 \setminus \bigcup_{a \in \mathfrak{C}} T_a \right| = 0. \quad (6.2)$$

Choose one tile T_{a_0} and its outer sphere

$$\Sigma = \{x : |x - a_0| = 1\}.$$

Fix $p \in \Sigma$ and put $u = p - a_0$. For $n \geq 1$, let

$$U_n = B\left(p + \frac{2}{n}u, \frac{1}{n}\right).$$

Every U_n lies outside T_{a_0} , is open, and has positive volume. By (6.2), choose $x_n \in U_n \cap T_{a_n}$ with $a_n \neq a_0$. Since $x_n \rightarrow p$ and $|x_n - a_n| < 1$, the centers a_n eventually lie in a fixed bounded set. Local finiteness lets us pass to a subsequence with one fixed center $a \neq a_0$. Thus $p \in \overline{T_a}$.

The point p is not in T_a . Otherwise some $B(p, \varepsilon)$ lies in T_a . Choose

$$0 < t < \min\{\varepsilon, 1 - \alpha\}.$$

Then $p - tu \in T_{a_0} \cap T_a$, contradicting disjoint interiors. Therefore every $p \in \Sigma$ belongs to the boundary of a neighboring tile.

If $p \in \Sigma \cap \partial T_a$, then $|p - a| \in \{\alpha, 1\}$, hence $|a - a_0| \leq 2$. Only finitely many centers satisfy this bound. Consequently Σ is covered by finitely many neighboring inner or outer boundary spheres.

Translate so that $a_0 = 0$. The intersection of Σ with a neighboring sphere $\{x : |x - a| = s\}$, $s \in \{\alpha, 1\}$, satisfies

$$|x|^2 = 1, \quad |x - a|^2 = s^2,$$

and hence

$$2a \cdot x = |a|^2 + 1 - s^2. \quad (6.3)$$

Because $a \neq 0$, (6.3) is a genuine plane. Its intersection with Σ is empty, a tangent point, or a circle, and therefore has zero surface measure on Σ . A finite union of such sets cannot cover Σ , whose area is 4π .

There is no coincident-sphere exception. A neighboring outer sphere equals Σ only when it has the same center, which is the original tile or a forbidden duplicate. A neighboring inner sphere cannot equal Σ because $\alpha < 1$. We have reached a contradiction. Scaling back proves Theorem 1.2.

A Logical dependency and evidence boundary

The proof has four logically distinct layers:

1. **Exact analysis:** operator decomposition, phase representation, strict counting, weighted-lattice identities, endpoint estimates, and analytic propagation.

2. **Finite mode analysis:** explicit radial and angular inventories and exact Weyl payments through k_{11} .
3. **Certified finite computation:** only the rectangle and $K = 200$ base predicates stated after Lemma 4.4.
4. **Elementary geometry:** the non-tiling proof, which uses none of the spectral work.

The legacy numerical Bessel experiments are not proof inputs. External Lorch inequalities [4] are used only for explicitly stated first-positive-zero bounds; the shell variational comparison, radial-index preservation, multiplicity counts, and Weyl payments are internal arguments. The audited environment reconstructed the exact specializations from the published statement but did not have access to the article’s full proof, so an external submission must either cite that theorem conventionally or replace those bounds by self-contained proofs. The relevant source cards are `sources/lorch_bessel_zeros.md`, `sources/round19_bessel_zero_bounds.md`, and `sources/round20_bessel_zero_bounds.md`.

All repository paths in this draft are relative to the repository root. They are printed as paths, rather than as external hyperlinks, because the audited snapshot at local commit `d89b609` has not yet been published. A review copy must therefore be distributed together with a repository checkout containing that commit; without the checkout, the internal records should be treated as unavailable. The final graph and validation records are `state/proof_obligations.yml`, `state/last_validation_report.md`, and `rounds/polya-main/round_022/reviews/derived-state-final-audit.md`.

B Work required before external submission

This draft is mathematically complete only relative to the listed detailed lemmas and certified supplements. Before external submission it should be expanded as follows:

1. move the full proofs of Propositions 2.1 and 2.2 into analytic appendices;
2. replace internal packet language in Lemmas 3.1–4.4 with conventional numbered propositions and complete proofs;
3. print the Round 17–20 mode inventories, zero exclusions, fixed and moving payments, and the corrected cap-74 argument;
4. provide a computer-assisted-proof supplement containing the exact algorithms, rational partitions, environment, hashes, digest, and replay instructions;
5. give conventional bibliographic citations for all external inputs, respecting their exact statement-level scopes;
6. run an independent human line-by-line reconstruction; and
7. perform a current literature comparison before using any novelty or priority language.

These tasks concern exposition, external verification, and scholarly scope. They do not broaden Theorem 1.1 to general domains.

References

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